BTEC Level 3 Applied Science · Unit 1 Chemistry · Worksheet

Covalent Bonding

Original JDScience practice. These questions were written for this resource and are not official exam-board items.

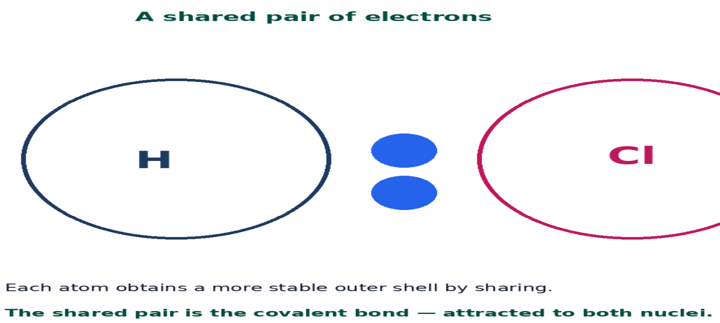
Learning focus: shared pairs, simple molecular versus giant covalent structures.

A Retrieval

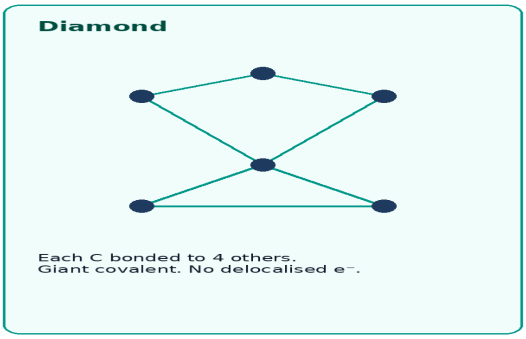
1. Define a covalent bond.

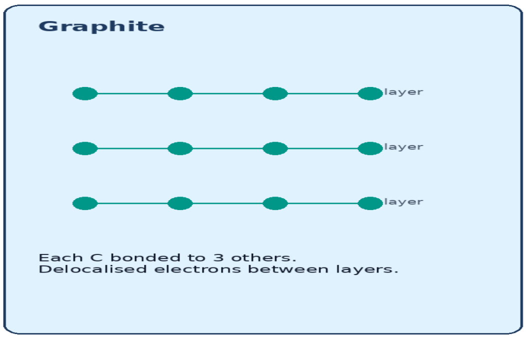
2. What is a lone pair?

3. Give one simple-molecular and one giant-covalent example.



4. Explain how a chlorine molecule is held together, referring to both nuclei.





5. Describe the bonding and structure of diamond.

6. Why does graphite conduct electricity when diamond does not?

C Application

7. Complete the table for the number of shared pairs.

Molecule	F₂	O₂	N₂	CH₄

8. Explain why water has a much lower boiling point than silicon dioxide.

9. Why does boiling water not break O–H covalent bonds?

D Exam-style practice

10. Describe how a covalent bond forms in a hydrogen molecule. [3]

11. Explain why iodine melts far below diamond. [4]

12. Explain why carbon dioxide does not conduct electricity. [2]

E Challenge

13. Ammonia can accept a proton to form NH_4^+. Explain this using a lone pair and a dative covalent bond.